Notes: Campello (JF, 2002)

Internal Capital Markets in Financial Conglomerates: Evidence from Small Bank Responses to Monetary Policy

Contents

1 Big picture

- **Research question.** Do internal capital markets inside financial conglomerates relax the external financing constraints faced by small bank affiliates? Do those internal markets allocate capital efficiently across affiliates?
- **Main idea.** A Fed tightening makes insured deposits and other external funds harder to use for loan expansion. Small stand-alone banks should become more dependent on their own cash flow. Small banks that belong to bank holding companies (BHCs) with at least one large bank may instead borrow internally from the unconstrained part of the conglomerate.
- **Empirical object.** The paper studies how the sensitivity of small-bank loan growth to operating income changes across monetary-policy regimes.
- **Main finding on constraints.** When monetary policy tightens, the loan growth of small independent banks becomes more tied to their own income. By contrast, small members of large BHCs become less tied to their own income, consistent with internal capital markets insulating affiliates from external funding shocks.
- **Main finding on monetary transmission.** Internal capital markets can weaken the bank-lending channel of monetary policy because BHC-affiliated small banks are less forced to contract lending after Fed tightenings.
- **Main finding on allocation efficiency.** Internal capital markets do not always allocate funds well. In small, constrained BHCs, poorly managed affiliates appear to receive support relative to well-managed affiliates. In large, unconstrained BHCs, internal funds appear to favor better affiliates.
- **Economic takeaway.** Internal capital markets can relax financing constraints, but whether they improve investment allocation depends on the constraints and governance of headquarters.

2 Incremental contribution of the paper

- **A cleaner setting for internal capital markets.** The paper studies banks rather than industrial conglomerate segments. Bank affiliates operate in similar businesses, report comparable data, and are organized partly because of legal and political restrictions on branching.
- **Exogenous funding shock.** Monetary policy creates time variation in the cost and availability of bank external funds. This provides a sharp way to test whether internal capital markets buffer affiliates from capital-market shocks.
- **Comparison of similar units.** The main contrast is between small stand-alone banks and small banks affiliated with large BHCs, not between unrelated divisions in very different industries.
- **Two questions in one setting.** The paper first asks whether internal capital markets exist and relax constraints. It then asks whether those markets allocate funds efficiently across affiliates.
- **Endogeneity checks.** The analysis uses location-size matched independent banks, parent capitalization, and premerger bank behavior to address the concern that BHC affiliates are selected firms with different lending behavior even before affiliation.

- **Connection to monetary policy.** The findings also speak to the bank-lending channel: conglomerate structure can affect how Fed policy is transmitted into credit supply.

3 Data and measurement

3.1 Bank-level data

The main data come from quarterly Call Reports for insured U.S. commercial banks. The core sample covers 1981:IV to 1997:II, because the key variables needed for the main tests can be consistently sampled over that period.

- **Final sample.** After excluding mergers, ownership changes, outliers, and banks without sufficient loan histories, the sample contains 601,858 bank-quarter observations.
- **Small and large banks.** Each quarter, banks in the top percentile of the asset distribution are classified as large, while banks below the 90th percentile are classified as small.
- **Small-bank categories.** Small banks are split into stand-alone banks, members of multibank BHCs without a large bank, and members of multibank BHCs with at least one large bank.
- **Main comparison.** The constraint-relaxation tests compare small stand-alone banks with small members of large BHCs.
- **Loan growth.** The investment variable is the change in log loans.
- **Internal funds.** The cash-flow variable is operating income, measured as net income relative to beginning-of-period loans.
- **Controls.** The first-stage regressions include lagged loan growth, nonperforming loans, capitalization, asset growth, state fixed effects, metropolitan-area status, and liquidity changes.

3.2 Monetary-policy variables

The second-stage tests use four monetary-policy proxies. All are transformed so that higher values represent tighter monetary policy.

- **Fed Funds.** The effective federal funds rate.
- **Paper-Bill.** The spread between six-month prime commercial paper and 180-day Treasury bills.
- **Strongin.** A reserve-based measure designed to isolate monetary-policy disturbances.
- **Boschen-Mills.** A narrative indicator that classifies the stance of monetary policy from strongly contractionary to strongly expansionary.

3.3 BHC and affiliate-efficiency variables

- **Parent capitalization.** A large BHC is classified as well-capitalized if at least one large bank in the BHC exceeds period-specific capital-to-asset cutoffs. Otherwise, it is classified as poorly capitalized.

- **Affiliate quality.** For efficiency tests, affiliates inside the same BHC are ranked by the change in nonperforming loans to total loans, adjusted by state averages.
- **Poorly and well-managed affiliates.** The paper compares affiliates in the worst quartile of within-BHC loan-performance changes with affiliates in the best quartile.
- **BHC-level data.** Consolidated BHC data from Y-9 Tapes are used to summarize small and large BHCs, including nonbank activities, public listing, inside ownership, and analyst coverage.

4 Methodology and empirical specifications

4.1 Step 1: quarterly loan-growth regressions

For each quarter and bank group, Campello estimates a cross-sectional loan-growth regression of the form

$$\begin{aligned}\Delta \log(Loans)_{i,t} = \alpha_t + \sum_{k=1}^{4} a_k \Delta \log(Loans)_{i,t-k} + b_1 Nonperforming_{i,t-1} \\ + b_2 Capitalization_{i,t-1} + b_3 \Delta \log(Assets)_{i,t-1} + \sum_{s} \gamma_s State_{s,i} \\ + b_4 Metro_i + b_5 \Delta \log(Liquidity)_{i,t} + \delta_t Income_{i,t-1} + \varepsilon_{i,t}.\end{aligned} \tag{1}$$

The coefficient of interest is δ_t, the loan-growth sensitivity to internal income in quarter t for a given bank group.

- A higher δ_t means loan growth depends more on the bank's own income.
- A lower δ_t means the bank is less constrained by its own internal funds.
- The liquidity variable is instrumented with lagged log liquidity because both lending and liquidity are bank decisions.

4.2 Step 2: monetary-policy response of cash-flow sensitivity

The first-stage coefficients are stacked into a time series and regressed on lagged monetary-policy variables:

$$\delta_t^g = a^g + \sum_{k=1}^{8} \phi_k^g MP_{t-k} + \Theta^g Z_t + u_t^g, \tag{2}$$

where g indexes the bank group, MP is a monetary-policy measure, and Z_t includes a time trend, seasonal quarter dummies, and in some specifications eight lags of real GDP growth.

The main statistic is the sum $\sum_{k=1}^{8} \phi_k^g$.

- For independent banks, a positive sum means tightening increases loan-income sensitivity.
- For BHC members, a zero or negative sum means affiliation reduces dependence on affiliate-level income during tightening.

- The paper uses Newey-West standard errors and SUR systems to test cross-group differences in the summed coefficients.

4.3 Robustness and endogeneity checks

- **Location-size matched sample.** Each small BHC affiliate is compared with a small independent bank in the same state and adjacent in the asset distribution.
- **Parent capitalization.** If internal capital markets work through access to external funds at the BHC level, affiliates of well-capitalized BHCs should be more insulated than affiliates of poorly capitalized BHCs.
- **Premerger behavior.** Independent banks that later merge into large BHCs are compared with banks that remain independent to see whether future BHC members already behave differently before affiliation.

4.4 VAR evidence on monetary-policy transmission

To connect internal capital markets to the bank-lending channel, the paper estimates separate VARs for small independent banks, small members of large BHCs, and large banks. The VAR includes real GDP, CPI, the Fed funds rate, deposits, liquid assets, and loans. It then traces the response of loans and liquid securities to a 100-basis-point Fed funds shock.

4.5 Efficiency tests inside BHCs

The allocation-efficiency tests compare relatively poorly and well-managed small affiliates within the same BHC. The idea is simple:

- Under **winner picking**, poorly managed affiliates should become more dependent on their own cash flow during Fed tightenings, while well-managed affiliates should be protected.
- Under **inefficient cross-subsidization**, poorly managed affiliates should be protected, while well-managed affiliates become more constrained.
- The tests are run separately for small BHCs and large BHCs to see whether headquarters' own financial constraints and monitoring environment matter.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether small bank affiliates with access to a large-bank BHC respond differently to monetary-policy shocks than otherwise similar small independent banks. The design is strongest for testing whether internal capital markets relax funding constraints during tight money.

5.2 Why monetary policy is useful

Monetary policy is a time-varying shock to banks' access to reserves and deposit funding. It affects all banks at the same time, but the effect should be more severe for small banks that face information frictions in uninsured funding markets.

5.3 Loan demand versus loan supply

A central concern is that monetary tightening reduces loan demand rather than loan supply. Campello addresses this in several ways: he compares small banks with similar balance sheets, includes local controls, adds GDP lags in the second stage, and uses BHC affiliation and parent capitalization to isolate the funding-supply channel.

5.4 Selection into BHCs

If future BHC affiliates already differ from independent banks, the results could reflect selection rather than internal capital markets. The paper's premerger tests find that banks that eventually merge into BHCs behave like other independent banks before the merger.

5.5 Interpretation of internal-capital-market efficiency

The efficiency results are not simply about whether money moves internally. They ask whether funds move toward the better affiliates when external finance becomes scarce. The answer depends on BHC structure: constrained small BHCs appear to subsidize weaker affiliates, while large BHCs appear to discipline weaker affiliates.

5.6 Limitations

The paper relies on observational variation in organizational form, even though the banking setting is cleaner than most segment-data settings. Affiliate quality is proxied by loan performance, which may capture both managerial quality and local economic shocks. The findings are most directly about U.S. banking conglomerates in the 1981-1997 regulatory and monetary-policy environment.

6 Tables and figures

6.1 Tables I and II: bank categories and first-step sensitivities

Read Table I:

- Table I reports balance-sheet statistics for small stand-alone banks, small banks in small BHCs, small banks in large BHCs, and large banks.
- The sample includes 413,568 stand-alone small-bank quarters, 102,398 small-BHC member quarters, 25,703 large-BHC member quarters, and 5,721 large-bank quarters.
- Small banks are much more deposit-financed than large banks and hold more liquid assets.

- Large banks rely less on deposits and hold a larger share of C&I loans.

Read Table II:

- Table II summarizes the first-step loan-income sensitivity δ_t by monetary-policy regime.
- In the overall sample, the mean sensitivity is 0.2160 for independent banks and 0.0853 for BHC members.
- During contractionary monetary-policy periods, independent banks' sensitivity rises to 0.2956, while BHC members' sensitivity falls to 0.0690.
- The contraction-period difference between independent banks and BHC members is 0.2266, with a reported p-value of 0.074.

Economic message: Small stand-alone banks look more exposed to funding constraints in tight-money periods. Small BHC affiliates are less dependent on their own income, consistent with support from internal capital markets.

Table I

Summary Statistics by Bank Category

This table presents the total number of bank-quarters in each of the four bank categories described in the text together with means and medians for (within-category) distributions of various balance-sheet data items (see definitions in the Appendix). The data are collected from the Call Reports over the 1981:IV to 1997:II period.

	Small Banks						Large Banks	
	Stand-alone Banks		Members of Small BHCs		Members of Large BHCs			
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Total assets (1992 $ millions)	69.9	47.1	92.8	66.8	120.2	98.0	23,652.7	11,607.2
% of banks located in MSAs	37.7	0.0	40.8	0.0	58.9	100.0	99.8	100.0
Liquid assets (% of total assets)	32.3	31.7	28.3	27.7	28.2	26.0	18.4	17.9
Total loans (% of total assets)	53.4	54.5	57.2	58.7	58.7	60.1	62.9	63.9
C&I loans (% of total loans)	19.0	16.1	19.5	17.0	22.4	20.4	33.5	32.8
Nonperform loans (% of total loans)	2.3	1.4	1.8	1.0	2.2	1.2	2.5	1.8
Total deposits (% of total assets)	88.4	89.2	88.6	89.6	87.1	90.1	71.5	73.7
Large CDs (% of total deposits)	10.8	8.7	10.7	8.7	11.5	9.0	15.8	12.7
Total equity (% of total assets)	9.5	8.9	8.5	8.1	7.6	6.9	6.4	6.1

Table I: Summary statistics by bank category.

Table II

Summary Statistics and Parametric Tests for First-Step δ_ts by Bank Category and across Monetary Policy Regimes

This table summarizes and compares the distributions of the estimated sensitivity of loan growtl to operating income funds for small independent banks versus small banks affiliated with BHC: with at least one bank ranked at the top percentile of the asset size distribution. Means for th overall sample are computed over the 1983:I to 1997:II period. Breakdown by the stance o monetary policy is based on the Boschen–Mills narrative measure, which is only available u until 1996:I. Tests for mean differences relax the assumption of equal variance across dat subsamples. Associated p-values are in parentheses.

First-Step δ_t	Overall Sample (58 Quarters) Mean (p-value)	Neutral or Expansionary Monetary Policy [Boschen–Mills $\geq$ 0] (34 Quarters) (A) Mean (p-value)	Contractionary Monetary Policy [Boschen–Mills < 0] (19 Quarters) (B) Mean (p-value)	Difference (A) − (B) Difference (p-value)
Independent banks	0.2160	0.1912	0.2956	−0.1044
	(0.000)	(0.000)	(0.000)	(0.070)
BHC members	0.0853	0.0921	0.0690	0.0231
	(0.191)	(0.291)	(0.563)	(0.875)
Differences:	0.1307	0.0991	0.2266	
Independents − BHC members	(0.132)	(0.310)	(0.074)	

Table II: First-step loan-income sensitivities by bank category and monetary-policy regime.

6.2 Table III: BHC affiliation and monetary-policy sensitivity

Read:

- Table III is the central constraint-relaxation table.
- The dependent variable is the estimated loan-income sensitivity from the first-step regressions.
- For independent banks, the sum of the monetary-policy coefficients is positive: tight money makes loan growth more dependent on internal income.
- For small members of large BHCs, the corresponding sums are negative in the unmatched sample.
- The difference between independent banks and BHC members is positive and statistically meaningful across the policy measures.
- Panel B repeats the comparison using location-size matched independent banks and finds similar or stronger differences.

Economic message: Affiliation with a large BHC changes how small banks respond to monetary-policy shocks. This is direct evidence that internal capital markets relax affiliate-level funding constraints.

Table III

Impact of Monetary Policy on Small Bank Loan Growth–Cash-Flow Sensitivity: Does BHC Affiliation Matter?

The dependent variable is the estimated sensitivity of loan growth to operating income for small independent banks versus small banks affiliated with BHCs with at least one bank ranked at the top percentile of the asset size distribution. All estimations are for bank-quarters ranked below the 90th percentile of the asset size distribution. In each estimation, the dependent variable is regressed on eight lags of a measure of monetary policy. All policy measures are transformed so that increases in their levels represent Fed tightenings. The sample period is 1981:IV to 1997:II (except for Paper–Bill (through 1997:I), and Boschen–Mills (through 1996:I)). In the bivariate regressions, eight lags of the log real GDP growth are added. The sum of the coefficients for the eight lags of the monetary policy measure are shown along with the p-values for the sum. Exclusion test rows report the p-values for the rejection of the hypothesis that the eight lags of the monetary policy measure do not forecast the sensitivity of loan growth to internal cash flows. Heteroskedasticity- and autocorrelation-consistent errors are computed with Newey–West lag window of size eight in each regression. The table displays (at the bottom) the differences between the sums of the eight lags of monetary policy of independent banks and BHC members, and the p-values for the test of the hypothesis that the sums are equal across categories. The standard errors for the difference of the sum of the coefficients are computed via a SUR system that estimates the bank categories regressions jointly.

		Monetary Policy Measure			
		Fed Funds	Paper–Bill	Strongin	Boschen–Mills
Panel A: Unmatched Samples					
Independent banks					
Univariate	Sum of coefficients	0.019	0.188	0.632	0.061
	Summation test (p-value)	0.025	0.003	0.013	0.307
	Exclusion test (p-value)	0.020	0.000	0.000	0.657
Bivariate	Sum of coefficients	0.022	0.223	0.658	0.025
	Summation test (p-value)	0.030	0.000	0.003	0.679
	Exclusion test (p-value)	0.012	0.000	0.000	0.577

Table III, part 1: BHC affiliation and monetary-policy effects.

BHC members					
Univariate	Sum of coefficients	−0.140	−0.888	−3.358	−0.600
	Summation test (p-value)	0.026	0.000	0.106	0.000
	Exclusion test (p-value)	0.137	0.001	0.003	0.003
Bivariate	Sum of coefficients	−0.217	−1.172	−3.584	−0.628
	Summation test (p-value)	0.021	0.000	0.099	0.000
	Exclusion test (p-value)	0.028	0.001	0.000	0.000
Differences: Independents − BHC members					
Univariate	Sum of coefficients	0.159	1.076	3.990	0.661
	Summation test (p-value)	0.034	0.021	0.040	0.044
Bivariate	Sum of coefficients	0.239	1.395	4.242	0.653
	Summation test (p-value)	0.023	0.016	0.065	0.070
Panel B: Location-Size Matched Samples					
Differences: Independents − BHC members					
Univariate	Sum of coefficients	0.189	1.466	4.702	0.518
	Summation test (p-value)	0.018	0.012	0.025	0.177
Bivariate	Sum of coefficients	0.274	1.774	4.858	0.448
	Summation test (p-value)	0.015	0.015	0.049	0.296

Table III, part 2: continuation and location-size matched sample.

6.3 Tables IV and V: parent capitalization and premerger comparisons

Read Table IV:

- Table IV splits small affiliates of large BHCs by whether the BHC is poorly or well capitalized.
- Members of poorly capitalized BHCs behave more like constrained banks: their loan-income sensitivity is often positive after tight policy.
- Members of well-capitalized BHCs show negative responses, consistent with better access to internal funding.
- The differences between poorly and well-capitalized BHCs are positive across all policy measures.

Read Table V:

- Table V compares independent banks that will later merge into large BHCs with independent banks that do not merge.
- Premerging independent banks respond to Fed tightenings in the same direction as nonmerging independent banks.
- The differences between premerging and nonmerging banks are not statistically strong.

Economic message: The affiliation result is not simply due to selection of inherently different small banks into BHCs. The parent BHC's own ability to access funds matters exactly as the internal-capital-market hypothesis predicts.

Impact of Monetary Policy on Small Bank Loan Growth–Cash-Flow Sensitivity: Does BHC Capitalization Matter?

The dependent variable is the estimated sensitivity of loan growth to operating income for banks affiliated with well- versus poorly capitalized BHCs with at least one bank ranked at the top percentile of the asset size distribution. All estimations are for bank-quarters ranked below the 90th percentile of the asset size distribution. In each estimation, the dependent variable is regressed on eight lags of a measure of monetary policy. All policy measures are transformed so that increases in their levels represent Fed tightenings. The sample period is 1981:IV to 1997:II (except for Paper–Bill (through 1997:I), and Boschen–Mills (through 1996:I)). In the bivariate regressions, eight lags of the log real GDP growth are added. The sum of the coefficients for the eight lags of the monetary policy measure are shown along with the *p*-values for the sum. Exclusion test rows report the *p*-values for the rejection of the hypothesis that the eight lags of the monetary policy measure do not forecast the sensitivity of loan growth to internal cash flows. Heteroskedasticity- and autocorrelation-consistent errors are computed with Newey–West lag window of size eight in each regression. The table displays (at the bottom) the differences between the sums of the eight lags of monetary policy of independent banks and BHC members, and the *p*-values for the test of the hypothesis that the sums are equal across categories. The standard errors for the difference of the sum of the coefficients are computed via a SUR system that estimates the bank categories regressions jointly.

		Monetary Policy Measure			
		Fed Funds	Paper–Bill	Strongin	Boschen–Mills
Members of poorly cap. BHCs					
Univariate	Sum of coefficients	0.071	0.476	−0.222	0.709
	Summation test (*p*-value)	0.059	0.165	0.823	0.070
	Exclusion test (*p*-value)	0.006	0.149	0.001	0.001
Bivariate	Sum of coefficients	0.055	0.314	−0.749	0.551
	Summation test (*p*-value)	0.243	0.551	0.510	0.118
	Exclusion test (*p*-value)	0.004	0.391	0.159	0.269
Members of well-cap. BHCs					
Univariate	Sum of coefficients	−0.628	−4.757	−15.917	−3.279
	Summation test (*p*-value)	0.117	0.151	0.136	0.065
	Exclusion test (*p*-value)	0.000	0.001	0.274	0.191
Bivariate	Sum of coefficients	−0.934	−5.312	−18.418	−3.499
	Summation test (*p*-value)	0.075	0.214	0.149	0.024
	Exclusion test (*p*-value)	0.005	0.005	0.226	0.053
Differences: Poorly − well-capitalized BHCs					
Univariate	Sum of coefficients	0.699	5.233	15.695	3.988
	Summation test (*p*-value)	0.066	0.061	0.099	0.070
Bivariate	Sum of coefficients	0.989	5.626	17.669	4.049
	Summation test (*p*-value)	0.065	0.114	0.090	0.104

Table IV: BHC capitalization and affiliate loan-income sensitivity.

Impact of Monetary Policy on Small Bank Loan Growth–Cash-Flow Sensitivity: Pre-Merger Comparisons

The dependent variable is the estimated sensitivity of loan growth to operating income for independent banks that do not merge versus banks that will merge in the 1981:IV to 1997:II period. All estimations are for bank-quarters ranked below the 90th percentile of the asset size distribution. In each estimation, the dependent variable is regressed on eight lags of a measure of monetary policy. All policy measures are transformed so that increases in their levels represent Fed tightenings. The sample period is 1981:IV to 1995:I. In the bivariate regressions, eight lags of the log real GDP growth are added. The sum of the coefficients for the eight lags of the monetary policy measure are shown along with the *p*-values for the sum. Exclusion test rows report the *p*-values for the rejection of the hypothesis that the eight lags of the monetary policy measure do not forecast the sensitivity of loan growth to internal cash flows. Heteroskedasticity- and autocorrelation-consistent errors are computed with Newey–West lag window of size eight in each regression. The table displays (at the bottom) the differences between the sums of the eight lags of monetary policy of independent banks and BHC members, and the *p*-values for the test of the hypothesis that the sums are equal across categories. The standard errors for the difference of the sum of the coefficients are computed via a SUR system that estimates the bank categories regressions jointly.

		Monetary Policy Measure			
		Fed Funds	Paper–Bill	Strongin	Boschen–Mills
Nonmerging independent banks					
Univariate	Sum of coefficients	0.012	0.105	0.401	0.094
	Summation test (*p*-value)	0.040	0.026	0.041	0.009
	Exclusion test (*p*-value)	0.015	0.018	0.000	0.019
Bivariate	Sum of coefficients	0.022	0.155	0.503	0.083
	Summation test (*p*-value)	0.004	0.002	0.001	0.027
	Exclusion test (*p*-value)	0.000	0.002	0.000	0.090
Premerging independent banks					
Univariate	Sum of coefficients	0.077	0.289	1.273	0.430
	Summation test (*p*-value)	0.029	0.288	0.453	0.072
	Exclusion test (*p*-value)	0.001	0.027	0.000	0.012
Bivariate	Sum of coefficients	0.090	−0.068	0.494	0.624
	Summation test (*p*-value)	0.185	0.908	0.794	0.006
	Exclusion test (*p*-value)	0.031	0.029	0.033	0.091
Differences: Nonmerging − premerging					
Univariate	Sum of coefficients	−0.065	−0.184	−0.872	−0.336
	Summation test (*p*-value)	0.275	0.689	0.689	0.381
Bivariate	Sum of coefficients	−0.069	0.223	0.010	−0.541
	Summation test (*p*-value)	0.393	0.693	0.997	0.194

Table V: Premerger comparisons for independent banks.

6.4 Figure 1 and Table VI: monetary-policy transmission and BHC structure

Read Figure 1:

- Figure 1 plots impulse-response functions after a 100-basis-point shock to the Fed funds rate.
- Panel A shows small independent banks: after tightening, loans fall more than liquid securities after a few quarters.
- Panel B shows small members of large BHCs: their loan response is much flatter and resembles large banks.
- Panel C shows large banks: their balance-sheet composition is relatively insulated from the shock.

Read Table VI:

- Table VI compares small and large BHCs in 1986:II and 1997:II.
- Large BHCs are much larger, have more nonbank activity, are more likely to be publicly traded, and have more analyst coverage.
- Analyst coverage is much higher for large BHCs, suggesting greater external monitoring and information production.

Economic message: Internal capital markets can reduce the effect of Fed policy on bank credit supply. The table also helps explain why large BHCs may allocate internal capital differently from smaller BHCs.

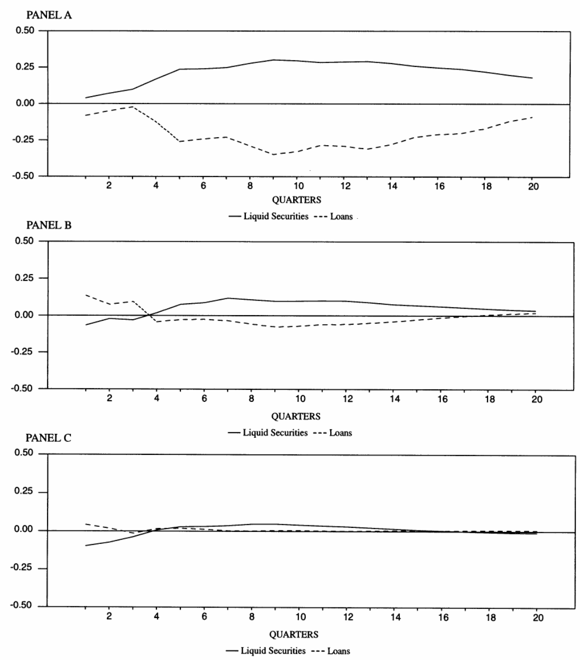

Figure 1. Small bank BHC-affiliation and the impact of monetary policy. Panel A: Small independent banks. Panel B: Small members of large BHCs. Panel C: Large banks. Impulse-response functions are estimated from a six-variable VAR that includes log real GDP, the change in log CPI, the Fed funds rate, bank deposits, liquid assets, and total loans (orthogonalization following this order). Four lags of each of the variables are used. Responses to a 100-basis-point shock to the Fed funds rate are computed separately for aggregate data on small independent

Figure 1: Impulse responses of liquid securities and loans to a Fed funds shock.

Table VI

Summary Statistics by BHC Category

This table presents the total number of subsidiary banks for each of the two BHC types (small and large) after merging Call Report and Y9 Tapes for the first and last quarters these two data sets overlap in the sample period used (1986:II and 1997:II, respectively). It also displays the proportion of BHCs' total assets and cash flows attributed to nonbank operations and the proportion of BHCs in the matched sample with data also available in CRSP. Inside ownership refers to the proportion of common stocks held by officers and directors in the BHCs totals (public BHCs only). Ownership information is collected from Compact Disclosure/Spectrum. The number of analysts covering the BHC (i.e., issuing earnings forecasts) is from I/B/E/S.

	1986:II				1997:II			
	Small BHCs		Large BHCs		Small BHCs		Large BHCs	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
Total assets (1992 $ billions)	1.5	0.9	25.2	13.8	1.8	0.7	75.4	44.0
Number of small bank affiliates	8.2	6.0	19.2	8.0	7.3	5.0	10.3	6.0
Nonbank assets (% of BHC total assets)	2.7	1.2	29.6	24.0	4.0	0.8	12.1	14.2
Nonbank income (% of BHC total income)	16.1	7.4	28.6	21.7	11.6	4.9	21.1	8.1
Number of BHCs	184		59		148		30	
Publicly traded (% of BHCs)	73.4		93.2		70.3		93.3	
Median inside ownership (% of shares)	6.7		2.2		7.3		2.1	
Median analyst coverage (number of analysts)	1		8		2		12	

Table VI: Summary statistics by BHC category.

6.5 Table VII: internal allocation and conglomerate structure

Read:

- Table VII compares poorly managed and well-managed affiliates within the same BHC.
- Panel A studies small BHCs. During monetary tightenings, poorly managed affiliates become less dependent on their own cash flow relative to well-managed affiliates.
- This pattern is consistent with inefficient cross-subsidization: weak affiliates receive support when internal funds are most valuable.
- Panel B studies large BHCs. Here the pattern reverses: poorly managed affiliates become more dependent on their own income, while well-managed affiliates are relatively protected.
- This pattern is consistent with winner picking or investment discipline inside large BHCs.

Economic message: Internal capital markets are not automatically efficient. They can relax constraints, but whether they fund the right affiliates depends on headquarters' own constraints and monitoring environment.

Table VII

Impact of Monetary Policy on Small Bank Loan Growth–Cash-Flow Sensitivity: Does Conglomerate Structure Matter?

The dependent variable is the estimated sensitivity of loan growth to operating income for relatively poorly and well-managed affiliates of the same BHC. All estimations are for bank-quarters ranked below the 90th percentile of the asset size distribution. In each estimation, the dependent variable is regressed on eight lags of a measure of monetary policy. All policy measures are transformed so that increases in their levels represent Fed tightenings. The sample period is 1981:IV to 1997:II (except for Paper–Bill (through 1997:I), and Boschen–Mills (1996:I)). In the bivariate regressions, eight lags of the log real GDP growth are added. The sum of the coefficients for the eight lags of the monetary policy measure are shown along with the p-values for the sum. Exclusion test rows report the p-values for the rejection of the hypothesis that the eight lags of the monetary policy measure do not forecast the sensitivity of loan growth to internal cash flows. Heteroskedasticity- and autocorrelation-consistent errors are computed with Newey–West lag window of size eight in each regression. The table displays (at the bottom) the differences between the sums of the eight lags of monetary policy of independent banks and BHC members, and the p-values for the test of the hypothesis that the sums are equal across categories. The standard errors for the difference of the sum of the coefficients are computed via a SUR system that estimates the bank categories regressions jointly.

		Monetary Policy Measure			
		Fed Funds	Paper–Bill	Strongin	Boschen–Mills
Panel A: Poorly versus Well-Managed Affiliates within Small BHCs					
Poorly managed affiliates					
Univariate	Sum of coefficients	−0.041	−0.448	−1.253	−0.559
	Summation test (p-value)	0.393	0.190	0.325	0.091
	Exclusion test (p-value)	0.913	0.014	0.408	0.084
Bivariate	Sum of coefficients	−0.116	−0.723	−1.503	−0.407
	Summation test (p-value)	0.071	0.038	0.416	0.468
	Exclusion test (p-value)	0.000	0.000	0.000	0.001
Well-managed affiliates					
Univariate	Sum of coefficients	0.120	0.816	3.158	0.445
	Summation test (p-value)	0.057	0.042	0.027	0.228
	Exclusion test (p-value)	0.107	0.000	0.000	0.415
Bivariate	Sum of coefficients	0.206	1.082	4.394	0.963
	Summation test (p-value)	0.011	0.061	0.025	0.024
	Exclusion test (p-value)	0.067	0.000	0.000	0.000

Table VII, part 1: poorly versus well-managed affiliates within small BHCs.

		Fed Funds	Paper–Bill	Strongin	Boschen–Mills
Differences: Poorly − well-managed					
Univariate	Sum of coefficients	−0.161	−1.264	−4.412	−1.005
	Summation test (p-value)	0.098	0.077	0.066	0.050
Bivariate	Sum of coefficients	−0.322	−1.806	−5.897	−1.370
	Summation test (p-value)	0.024	0.030	0.029	0.024
Panel B: Poorly versus Well-Managed Affiliates within Large BHCs					
Poorly managed affiliates					
Univariate	Sum of coefficients	0.206	1.716	6.630	0.156
	Summation test (p-value)	0.001	0.000	0.000	0.724
	Exclusion test (p-value)	0.028	0.000	0.000	0.000
Bivariate	Sum of coefficients	0.281	1.884	6.118	0.111
	Summation test (p-value)	0.009	0.001	0.000	0.826
	Exclusion test (p-value)	0.008	0.000	0.000	0.015
Well-managed affiliates					
Univariate	Sum of coefficients	−0.048	−0.796	−2.891	0.304
	Summation test (p-value)	0.299	0.034	0.017	0.411
	Exclusion test (p-value)	0.000	0.000	0.202	0.081
Bivariate	Sum of coefficients	−0.149	−1.514	−4.566	−0.091
	Summation test (p-value)	0.034	0.002	0.012	0.832
	Exclusion test (p-value)	0.000	0.000	0.000	0.019
Differences: Poorly − well-managed					
Univariate	Sum of coefficients	0.254	2.512	9.521	−0.148
	Summation test (p-value)	0.066	0.008	0.009	0.841
Bivariate	Sum of coefficients	0.430	3.399	10.683	0.202
	Summation test (p-value)	0.046	0.006	0.014	0.825

Table VII, part 2: continuation and large-BHC comparison.

7 Short summary

- Campello studies internal capital markets in financial conglomerates using small-bank responses to monetary policy.
- The empirical setting is useful because banks report comparable affiliate-level data and monetary policy creates external funding shocks.
- The core sample is quarterly U.S. Call Report data from 1981:IV to 1997:II.
- The first-stage regression estimates the sensitivity of loan growth to operating income for different bank groups.
- The second-stage regression asks whether that sensitivity rises after monetary-policy tightening.
- Small independent banks become more cash-flow sensitive when the Fed tightens.
- Small members of large BHCs become less cash-flow sensitive during tight money.
- The result is stronger when the parent BHC is well capitalized and disappears when the parent is poorly capitalized.
- Banks that later merge into BHCs do not already look different before the merger, reducing the selection concern.
- VAR evidence shows that BHC-affiliated small banks look more like large banks than like independent small banks after Fed funds shocks.
- In small BHCs, internal capital markets appear to subsidize weak affiliates.
- In large BHCs, internal capital markets appear to favor better affiliates.
- The paper's final interpretation is that frictions between headquarters and external capital markets help explain when internal capital markets become inefficient.

8 Conclusion

8.1 Core conclusion

The paper shows that internal capital markets in financial conglomerates relax the financing constraints faced by small bank affiliates. When monetary policy tightens, small independent banks become more dependent on their own income, but small banks affiliated with large BHCs are insulated from the tightening.

8.2 Economic intuition

A small stand-alone bank has limited access to cheap uninsured external funds. A small affiliate in a large BHC can rely on funds raised or reallocated by the broader organization. This internal funding channel breaks the tight link between the affiliate's own income and its loan growth.

8.3 Contribution revisited

The paper improves on earlier internal-capital-market studies by using a cleaner institutional setting, comparable affiliate-level data, and monetary-policy shocks to external finance. It also moves beyond the existence of internal capital markets and studies allocation efficiency within conglomerates.

8.4 Implications

The results imply that conglomerate organization can weaken monetary-policy transmission through the bank-lending channel. They also imply that internal capital markets can be valuable for relaxing constraints but may distort allocation when headquarters itself is constrained or weakly monitored.

8.5 Limitations

The setting is specific to banking and to the 1981-1997 period. The efficiency test relies on loan performance as a proxy for affiliate quality. The paper cannot fully rule out every source of unobserved loan-demand heterogeneity, although the design and robustness tests substantially reduce this concern.

8.6 Takeaway

Internal capital markets matter because they change how firms respond to external financing shocks. But their value depends on governance: they can either fund the right affiliates or protect the wrong ones.